🛠️ IT Support Careers and Certifications

Fulllion Creative Works

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1. The Entry-Level IT Path

Most IT careers start by helping people with their computers, then move toward running the systems behind them. Titles vary by company, but the path usually looks like this:

Stage	Common titles	What you do
1. Help desk	Help desk technician, IT support specialist, service desk analyst, Tier 1 support	Answer tickets, calls and chats. Reset passwords, fix printers and email, set up accounts, and escalate harder problems. Customer service matters as much as technical skill.
2. Desktop support	Desktop support technician, field technician, Tier 2 support	Hands-on work: set up and repair PCs and laptops, install software, replace hardware, image machines, and fix problems Tier 1 couldn't.
3. Systems administration	Junior system administrator, IT administrator	Manage servers, user accounts and permissions (often Microsoft Active Directory and Entra ID), backups, updates and scripts.
4. Specialize	Network administrator or engineer, security analyst, cloud engineer, DevOps engineer	Go deep in one area: **networking** (Network+, then Cisco CCNA), **security** (Security+, then more), or **cloud and automation** (see Getting Started with DevOps).

You don't have to follow every step, and many people move faster with a strong homelab and the right certificate. But help desk experience teaches troubleshooting and dealing with people under pressure, which every later role needs.

A realistic note on the job market. The US Bureau of Labor Statistics (BLS) projects employment of computer support specialists to **decline 3%** from 2025 to 2035, but still expects about **48,700 openings a year**, mostly from people moving into other jobs or retiring. It reports a median pay of $62,890 in May 2025, and notes that some employers accept a high school diploma plus relevant IT certifications. These are US figures; check your own country's sources. In short: entry-level IT is competitive, so practice and certifications help you stand out.

2. Do You Need Certifications?

Not always, but they often get your resume past the first filter, especially without a degree or experience.

- **CompTIA says A+ appears in more tech support job listings than any other IT credential.** That's CompTIA's own claim, but you can test it yourself: search a job site in your area for "help desk" or "IT support", and note which certifications the postings mention.
- **Some US government and defense roles require specific certifications.** CompTIA lists A+ and Security+ as approved for several US Department of Defense (DoD 8140) work roles.
- **The Google certificate is a training program;** A+ is an industry exam. They do different jobs, and work well together (Section 3).

3. Which Certificate First?

Take both if you can: they're what a lot of job postings look for.

	Google IT Support Professional Certificate	CompTIA A+
What it is	A self-paced online course on Coursera, with a certificate when you finish	Two proctored exams; you choose how to study
Teaching included?	Yes: videos, labs and quizzes	No; the exam fee is just the test (CompTIA sells training separately, and free resources exist)
Time	About 3 to 6 months, or 3 months at 10 hours a week	Depends on you; many people study for a few months
Cost	A monthly subscription, so finishing faster costs less	One exam voucher for each of the two exams
Best for	Complete beginners who want structure	Proving your knowledge to employers with the industry-standard exam

A good plan: take the Google certificate to learn, then use free resources (Section 9) to fill the gaps and pass A+. Coursera says the Google program prepares you for the A+ exams, and Google calls the pair a "dual credential".

4. Google IT Support Professional Certificate

Made by Google and taught on Coursera. As of November 2026:

- **Price:** Google's certificate page lists **$49 USD a month in the US and Canada after a 7-day free trial**. Prices may be lower in other countries, Coursera offers **financial aid** to those who qualify, and it's also included in a **Coursera Plus** subscription. **Check current pricing** before you enroll.
- **Time:** Coursera estimates **3 months at 10 hours a week**; Google says 3 to 6 months. It's self-paced, so you can go faster or slower.
- **Level:** Beginner; no experience or degree needed.
- **What it covers:** day-to-day IT support tasks such as computer assembly, wireless networking, installing programs and customer service; troubleshooting from first report to fix; and Linux, DNS, the command line and binary. Hands-on labs cover troubleshooting, networking, operating systems, system administration, automation and security. The courses are:
  1. Technical Support Fundamentals
  2. The Bits and Bytes of Computer Networking
  3. Operating Systems and You: Becoming a Power User
  4. System Administration and IT Infrastructure Services
  5. IT Security: Defense against the digital dark arts
  6. Accelerate Your Job Search with AI (optional; included in all Google Career Certificates)
- **College credit:** the program is **ACE recommended for up to 15 college credits**, the equivalent of 5 bachelor's-level courses, at participating US colleges. Each college decides whether to accept them.
- **A+ discount?** Neither Google's nor Coursera's page mentioned an A+ exam discount in November 2026. Both describe the program as **preparation** for the A+ exams. Check the program's own pages for any current offers.
- **Jobs:** Coursera says graduates can apply directly to jobs with Google and more than 150 US employers.

5. CompTIA A+

CompTIA A+ is the standard entry-level IT certification. It's vendor-neutral, so it covers Windows, macOS, Linux, mobile devices and hardware from any maker.

5.1 The current version

- **Current series: V15**, exams **220-1201 (Core 1)** and **220-1202 (Core 2)**, launched March 25, 2025.
- **The old V14 series (220-1101 and 220-1102) retired** in English on September 25, 2025. Don't buy V14 study materials.
- **V15 is expected to retire around 2028,** about three years after launch, as CompTIA exams usually do.
- **You need to pass both exams,** in any order, **from the same version.** You can't mix a V14 exam with a V15 exam.

5.2 The exams

	Core 1 (220-1201)	Core 2 (220-1202)
Questions	Up to 90: multiple choice, drag and drop, and performance-based tasks	Same
Time	90 minutes	90 minutes
Passing score	675 (on a scale up to 900)	700 (on a scale up to 900)
Topics	Mobile devices (13%), networking (23%), hardware (25%), virtualization and cloud (11%), hardware and network troubleshooting (28%)	Operating systems (28%), security (28%), software troubleshooting (23%), operational procedures (21%)

CompTIA recommends **12 months of hands-on experience** in an IT support role. You don't need it to take the exams, but a homelab (Section 11) is the next best thing.

Cost: you buy one exam voucher per exam, so two in total. **Check current pricing** on CompTIA's site, and watch for bundles that include a retake.

5.3 Keeping it current

A+ is valid for **three years**. To renew, you can:

- earn **20 continuing education units (CEUs)** during the three years, from activities such as training, other IT certifications, work experience or CompTIA's **CertMaster CE** course; or
- earn a **higher-level CompTIA certification**, such as Network+ or Security+, which renews A+ as well.

A continuing education fee also applies; check CompTIA's continuing education pages for the current amount and rules.

6. Hardware Basics: RAM, and Why More Isn't Always Faster

Memory upgrades are one of the most common jobs in desktop support, and A+ Core 1 tests them. The basics:

- **Capacity (GB)** is how much the PC can hold in memory at once. If you run out, Windows moves data to the much slower drive (the page file), and everything slows down. Once you have enough for your workload, extra capacity mostly sits unused.
- **Speed** is the data rate, in MT/s (often labeled MHz), together with timings such as **CL** (lower is quicker). Your motherboard and processor set the maximum.
- **Generation** matters: DDR4 and DDR5 aren't interchangeable, and each only fits boards made for it.
- **Dual channel:** two matching modules in the right pair of slots give the processor two paths to memory at once. Which slots count as a pair varies by motherboard (often the second and fourth), so check the manual. One large stick can be slower than two smaller ones.
- **Buy matched kits.** Kingston, a RAM maker, warns that mixing modules, even with the same part number, can cause lower performance, instability or incompatibility. Kits are sold as matched pairs or sets.

- **Four sticks aren't automatically faster.** Kingston also notes that installing four modules doesn't give "quad channel" on most desktop processors, and filling every slot can lower the speed the system can run at.
- **Turn on XMP or EXPO.** Performance RAM starts at a safe default speed; Intel says XMP modules first boot at standard JEDEC settings. To get the speed on the box, turn on **XMP** (Intel) or **EXPO** (AMD) in the BIOS.

Check what a PC has. In Windows, open **Task Manager** > **Performance** > **Memory** to see the speed and how many slots are used. Or run this in PowerShell to list each module:

```
Get-CimInstance Win32_PhysicalMemory |
  Format-Table BankLabel, DeviceLocator, Capacity, Speed, PartNumber
```

Worth watching: OCmodshop's **"Here's Why Your RAM Upgrade Backfired"** swaps between 16, 32 and 64 GB of DDR4-2666 in a gaming laptop with an 8 GB RTX 2070, and runs ten games three times each, logging load times, RAM and VRAM use, and frame rates. What it found:

- **Dual channel mattered more than capacity.** The one configuration that ran in single channel, 32 GB, was the slowest in several games, despite having more memory than the 16 GB setup. In Cyberpunk 2077 it averaged about 45 frames per second, against about 61 for 16 GB in dual channel, and two other games lost over 20%.
- **Extra capacity mostly helped loading.** 64 GB shaved a few seconds off some load times, but frame rates in dual channel were close to identical from 16 to 64 GB.
- **16 GB can run close to full.** In some games, Windows and the game used nearly all 16 GB. The closer you get to the limit, the more the system leans on the page file, and the more likely stutters are.
- **Its verdict:** 16 GB is still fine for many games if you aren't running much in the background; 32 GB gives headroom for a new build or for streaming; and if a PC has a single stick, adding a matching second one is a cheap, large improvement.
- **Its limits:** one laptop, a dedicated graphics card, and DDR4. PCs with integrated graphics (APUs) share system RAM with the GPU, so results there can differ, as the video itself notes.

The lesson for support work: find out what the user actually runs, and check how the memory is installed, before recommending an upgrade. For "add a matching stick", remember Kingston's warning above: even identical modules aren't guaranteed to work together, so a matched kit is the safer buy.

7. Next Certifications: Network+ and Security+

After A+, CompTIA's next two certifications are the usual next steps. CompTIA describes A+ as building toward both.

CompTIA Network+

- **Current version:** V9, exam **N10-009**, launched June 20, 2024. CompTIA estimates it will retire in 2027.
- **One exam:** up to 90 questions, multiple choice and performance-based, in 90 minutes.

- **Covers:** networking concepts, setting networks up, running them, securing them, and troubleshooting them.
- **Leads to:** network administrator and network support roles, and Cisco's CCNA (Section 8).

CompTIA Security+

- **Current version:** V7, exam **SY0-701**, launched November 7, 2023. It retires in English on **June 11, 2027**.
- **New version:** V8, exam **SY0-801**, launches on **November 17, 2026**, with more on AI-related risks, security operations and modern environments. If you're starting from scratch now, study for V8, or book V7 well before its retirement date with V7 materials.
- **One exam:** up to 90 questions, passing score 750 (on a scale of 100 to 900).
- **Recommended experience:** Network+ and two years in a security or systems administrator role.
- **Where it shows up:** CompTIA lists Security+ as approved for many US DoD 8140 work roles, including system administrator, network specialist and incident responder, so it often appears in US government and contractor job postings.

Which jobs mention them? Rather than trust anyone's statistics, search your local job sites for the roles you want, and count which certifications the postings list. That tells you what employers near you actually ask for.

8. Vendor Certifications

Vendor certifications prove skill with one company's products. Choose the ones that match the jobs near you.

Certification	From	Good for	Notes (as of November 2026)
Cisco Certified Support Technician (CCST) IT Support	Cisco	Help desk	Exam 100-140, 50 minutes. Covers IT support tasks, hardware, connectivity, OS and app issues, threats, and tools. Cisco also offers CCST Networking and Cybersecurity.
CCNA	Cisco	Networking	Exam 200-301 (v1.1). No prerequisites, but best after Network+ or equivalent study.
Microsoft 365 Certified: Endpoint Administrator Associate	Microsoft	Desktop support and sysadmin in Microsoft workplaces	Exam MD-102. Intermediate. Managing devices with Intune, Windows Autopilot and Entra ID. Renews every 12 months with a free online assessment.
Microsoft Certified: Azure Fundamentals	Microsoft	A first step toward cloud	Exam AZ-900. Beginner.
AWS Certified Cloud Practitioner	Amazon	A first step toward cloud	Foundational. 90 minutes, valid for three years.
CompTIA Linux+	CompTIA	Linux administration	V8, exam XK0-006, launched July 15, 2025.

Certification	From	Good for	Notes (as of November 2026)
ITIL Foundation	PeopleCert	Service desk processes (tickets, incidents, changes)	Often mentioned in service desk postings. Training providers report a new ITIL Version 5 in 2026; check PeopleCert for which version to take.

9. Free Study Resources

Video courses

- **Professor Messer** (free videos, with optional paid notes and practice exams):
  - CompTIA A+ 220-1201 (Core 1): 63 videos, about 10 hours.
  - CompTIA A+ 220-1202 (Core 2): 74 videos, about 13.5 hours.
  - CompTIA Network+ N10-009 and Security+ SY0-701.
  - He also posts recordings of his study group sessions on the site.

Courses and labs

- **Cisco Networking Academy** has free courses. Its free **Getting Started with Cisco Packet Tracer** course is how you download **Packet Tracer**, Cisco's free network simulator, with a free account.
- **Microsoft Learn** has free training paths for every Microsoft certification, plus free practice assessments for many exams.
- **Linux Journey** teaches the Linux command line for free, with no sign-up.
- **Introduction to Linux (LFS101)** from the Linux Foundation is free and self-paced.

Roadmaps

- roadmap.sh has free, visual roadmaps for Linux, Cyber Security, DevOps and more.

Practice exams: use one reputable set of practice questions near the end of your studies, to find weak spots. Avoid "brain dump" sites that sell real exam questions: using them breaks the exam rules and can cost you your certification.

10. Toward Security: Protect Your SSH Keys

If you head toward security, you'll learn how attackers think, so you can defend against them. Here's one small, practical example.

SSH keys let you sign in to servers without a password. The private key file can be protected with a **passphrase**. Keon Bowman's short **"ssh2john in 60 Seconds"** shows why that passphrase matters. Starting with a copied private key that asks for a passphrase, it:

1. converts the key into a hash that **John the Ripper** can work on (the job of the **ssh2john** script);
2. runs John against the hash with **rockyou**, a well-known list of real passwords from an old data breach;
3. recovers a short, common passphrase in seconds, then uses the key to sign in to the server as that user.

All the guessing happens offline, on the attacker's own machine, so the server never sees a failed login. The video itself says to do this only on systems you have explicit authorization to test.

The defensive lesson:

1. **Always set a passphrase** on SSH private keys, and make it long and unique. Anything short, or anything that has ever appeared in a breach list, falls just as fast as the one in the video. Windows 10 and 11 include the OpenSSH client. To make a new key, run this and enter a passphrase when asked:

   ```
   ssh-keygen -t ed25519
   ```

2. **Add or change the passphrase** on an existing key:

   ```
   ssh-keygen -p -f $env:USERPROFILE\.ssh\id_ed25519
   ```

3. **Keep private keys private.** Never email them, commit them to Git, or copy them to shared folders. The `.pub` file is the one you share.

Want to practice attacks legally? Use platforms built for it, where you have permission:

- **TryHackMe** has a free tier with free rooms and limited access to its learning paths, aimed at beginners.
- **Hack The Box** releases a new machine to hack for free every week, with more content on paid plans.
- **OverTheWire** wargames, starting with Bandit, are free.

Never test these techniques on systems you don't own or have written permission to test.

11. Your Homelab Practice List

A **homelab** is a practice setup at home, where you can break things and fix them without risk. It's also great hands-on practice for A+ topics, and something to talk about in interviews.

11.1 What you need

- **Virtual machines (VMs)** on your own PC, so you can install and wreck operating systems freely:
  - **Hyper-V** is built into Windows 10 and 11 **Pro or Enterprise** (not Home). Turn it on from an administrator PowerShell window, then restart:

    ```
    Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
    ```

  - **VirtualBox** is free and works on Windows Home too.
  - **Windows Sandbox** (Windows Pro, Enterprise and Education, not Home) is a throwaway copy of Windows that's wiped when you close it: ideal for testing an unknown program. Turn it on from an administrator PowerShell window:

    ```
    Enable-WindowsOptionalFeature -Online -All `
      -FeatureName "Containers-DisposableClientVM"
    ```

- **An old PC or laptop,** for real hardware practice: opening it up, swapping RAM and drives, and reinstalling.
- **A spare USB flash drive** of 8 GB or more, for installers.

11.2 Practice with this site's guides

Each of these is a real task help desk and desktop support staff do, and each maps to A+ objectives.

Practice task	Guide	A+ topics it covers
Make a Windows install USB, enter the BIOS, install and activate Windows (do it in a VM first)	Windows Install Guide	Core 2: installing and upgrading operating systems. Core 1: boot settings and firmware.
Repair Windows with DISM, SFC and CHKDSK	DISM, SFC, and CHKDSK Windows Repair Guide	Core 2: Windows command-line tools and software troubleshooting
Close a stuck Windows Update reminder safely, schedule and pause updates, and fix an update that keeps failing	Fix Stuck Windows Update Prompts	Core 2: Windows Update settings, software troubleshooting, and recognizing fake security alerts
Move a Windows install to new hardware: check the license, set the BIOS boot and storage modes, and reactivate (practice with a VM or a spare drive)	Move Windows to a New PC	Core 2: installing operating systems and troubleshooting boot problems. Core 1: firmware settings
Move files off an old Windows drive, fix permissions, and remove old partitions	Remove an Old Windows Install and Keep Your Files	Core 2: file systems, Disk Management, permissions. Core 1: storage.
Install Linux alone or alongside Windows, then update it and check drivers	Linux Install Guide	Core 2: Linux basics, partitions and dual boot
See what a PC is connecting to, read the DNS cache, and match connections to programs	What Is Your PC Connecting To?	Core 1: networking tools. Core 2: security and malware symptoms.
Follow a malware removal process, from isolating the PC to securing accounts (practice the steps, not on a real infection)	Windows Malware Removal Guide	Core 2: malware removal steps, security, and operational procedures
Use `ipconfig`, `ping`, `tracert`, `nslookup`, `netstat` and other commands in Command Prompt, PowerShell and Bash	CLI Cheat Sheet	Core 1: network troubleshooting. Core 2: Windows and Linux commands.

11.3 More homelab projects

Pick a few that interest you. Each one teaches networking, storage, users and troubleshooting in a way you can explain in an interview.

1. **A home media server and personal cloud.** Madeline and Stephen's short video **"How To Make Your Own Streaming Service At Home!"** is a follow-up to their earlier video on copying their DVD collection onto a laptop. Here they recommend a dedicated always-on box instead, and show theirs: a Ugreen NASync DXP4800 network-attached storage (NAS) unit with two 8 TB hard drives. Points worth taking from it:

- **Two drives, mirrored.** The drives are cloned (RAID 1), so one can fail without losing anything, at the cost of half the space. They could unlink them for 16 TB, with no protection.
- **More than movies.** Through Ugreen's own app, they reach their files and photos from anywhere, much like a personal iCloud.
- **A cheaper route.** They suggest a used Mac mini left running without a monitor, plus one large drive, does the same job for much less.
- **Prices move.** They bought in September 2025 and note that drive and memory prices had risen since, so check current prices.

The video doesn't name a media server app beyond Ugreen's own. Two popular, vendor-neutral options:

- **Jellyfin** is free, open-source software (GPL license), with no paid tier. It's run by volunteers and funded by donations.
- **Plex** is a polished alternative with a free tier. Since April 29, 2025, streaming your own videos **away from home** needs a Plex Pass or Remote Watch Pass subscription; streaming at home doesn't. Check Plex's current plans.

Getting started with either one:

1. **Install the server** on the always-on machine. Jellyfin has a Windows installer, Linux packages, a Docker image (`jellyfin/jellyfin`) and Synology and TrueNAS options. Plex Media Server is a free download from plex.tv. Many NAS units, Ugreen's included, can run either through Docker.
2. **Organize your files the way the app expects**, so it finds the right titles and artwork. Both use the same pattern:

   ```
   Movies/Film Name (2019)/Film Name (2019).mkv
   TV Shows/Show Name (2018)/Season 01/Show Name (2018) - s01e01.mkv
   ```

   Spell out "Season 01"; don't shorten it to "S01". Put specials in "Season 00".
3. **Open the web page** the server shows after install (Jellyfin uses port 8096, for example `http://localhost:8096`). Create your account, then add a library for each folder (Movies, TV Shows).
4. **Install a client app** on your TV, phone or console. Jellyfin has apps for Android, Android TV and Fire TV, iOS, Roku, LG, Samsung and Xbox, and the web page works in any browser.
5. **Optional: hardware transcoding.** If a device can't play a file's format, the server converts it on the fly. An Intel CPU with Quick Sync or an NVIDIA GPU makes that much lighter. Jellyfin supports both for free, though not when Jellyfin runs in Docker on Windows or macOS. On Plex, according to its support article on hardware-accelerated streaming:
   - **Plex Pass required.** It needs an active Plex Pass, except on NVIDIA SHIELD and the WD My Cloud Pro PR2100 and PR4100, where it's on by default for everyone.
   - **Intel Quick Sync:** a 2nd-generation Intel Core (Sandy Bridge, 2011) or newer, with 5th-generation (Broadwell) or newer recommended. Encoding to HEVC needs a 7th-generation

Core or newer and Plex Media Server 1.41.3 or later. A NAS needs a Jasper Lake or better CPU to transcode to HEVC reliably.

- **NVIDIA** cards are supported (on Linux, with driver 450.66 or newer). GeForce cards can have a driver limit on how many videos they encode at once; Intel has no artificial limit. AMD GPUs are supported only "as is".
- **Not in virtual machines.** In Docker it works with extra setup, by passing the device through to the container. For Intel, Plex's pms-docker README adds `--device=/dev/dri:/dev/dri` to `docker run`. For NVIDIA, install the NVIDIA Container Toolkit and start the container with `--gpus all` (or `--runtime=nvidia`); its docs say the `video` driver capability is what loads the video codec libraries.
- **To turn it on:** in Plex Web, open **Settings** > **Server** > **Transcoder**, click **Show Advanced**, turn on **Use hardware acceleration when available**, and click **Save Changes**. No restart is needed.
- **To check it's working:** force a conversion (for example, pick 480p in the player), open the **Dashboard**, and look for "(hw)" next to the video format.
- **The trade-off:** Plex notes that hardware encoders are faster but can look slightly blurrier or blockier than software encoding, especially below 720p.

Community tips (not official Plex guidance). These come from user forums and reviews, not from Plex:

- Many users treat **Intel Quick Sync** as the best default for a Plex server: low power, no extra card, and many streams at once. Plex's own page supports the last point, since Intel has no artificial limit.
- Users report that Plex picks Quick Sync over an NVIDIA card when a PC has both, which matches the order in Plex's tech specs.
- For **4K HDR tone mapping**, XDA recommends a 12th-generation or newer Intel CPU. Plex's own minimums are lower.
- An **NVIDIA card** is a common choice when the CPU has no integrated graphics, such as an AMD Ryzen without graphics or an older Intel CPU. It draws more power, and needs a free PCIe slot and the right power connectors.
- **The GeForce limit:** older posts say GeForce cards can encode only 2 videos at once. NVIDIA's Video Encode and Decode GPU Support Matrix now lists current GeForce cards at **12 concurrent sessions**. Users often suggest workstation cards such as the T600 to avoid the limit, but the matrix lists the T400, T600 and T1000 at 8 sessions; higher-end cards such as the RTX A2000 are unrestricted. Check the matrix for your card. Some users run patched drivers to remove the limit; that's an unofficial modification, and this guide doesn't recommend it.

Two cautions the video skips. First, **a mirror isn't a backup.** RAID 1 survives a dead drive, but a deleted file, ransomware or a fire takes out both copies at once. Follow the **3-2-1 rule**: three copies, on two kinds of storage, with one off-site (project 6 below). Second, **copying discs you own can still be illegal.** In the US, breaking a DVD's copy protection falls under Section 1201 of copyright law, and personal backups aren't among its exemptions. Check the law where you live, or stick to media without copy protection.

2. **A local-first smart home.** Some smart home hubs now run on hardware you own, instead of sending everything to a company's cloud. Salem Techsperts' short on the Ugreen HomeAgent is a promotional piece filmed at Ugreen's launch event: it shows lights controlled by voice with no internet connection, and camera footage recognized and stored locally, and it ends with a link to reserve one. Treat its opening claim about cloud AI and your town's water supply as a sales pitch rather than a measured fact. Ugreen announced HomeAgent at IFA in September 2026, with pre-orders and a crowdfunding campaign; as of November 2026 it hadn't shipped or been independently reviewed, and Ugreen says some features, such as remote access, notifications and updates, may still need the internet. For a vendor-neutral option that exists today, **Home Assistant** is open source, keeps your data local, works with devices from many brands, and is run by the non-profit Open Home Foundation. It runs on its own Home Assistant Green hub or on hardware you already have.
3. **An offline knowledge server.** **Project NOMAD** is free, open-source software (Apache 2.0 license) from Crosstalk Solutions that turns a PC into an offline library. In The Modern Rogue's video **"What's In Project NOMAD: The Off-Grid Internet for the Apocalypse?"**, its creator, Chris Sherwood, explains how it works:
    - **It bundles existing open-source projects** rather than reinventing them: Wikipedia and thousands of reference books and manuals through Kiwix, Khan Academy courses through Kolibri, offline maps, and a local AI assistant through Ollama that can also search documents you upload. Each piece runs as a **Docker container**, and its "Supply Depot" catalog adds more, including any container you choose (the demo box runs Jellyfin with public domain films).
    - **Install Linux, then run one command.** In the video, he says it needs an ordinary x86 PC, such as an old laptop, mini PC or gaming rig, not ARM hardware like a Raspberry Pi or an Apple Silicon Mac. The project's GitHub page recommends Ubuntu or Debian.
    - **Local AI is the demanding part.** The demo box uses a small 1-billion-parameter model, which runs on a mini PC but visibly gets things wrong on camera, a good reminder that AI answers need checking. All of Wikipedia takes about 126 GB, and making a document searchable by the AI takes several times its size in extra disk space. The project's own page lists a minimum of 4 GB of RAM without the AI, and suggests 32 GB and a graphics card such as an RTX 3060 for the AI assistant.
    - **The portable "Nomad deck" in the video** pairs a mini PC with a battery (about four to five hours) and a travel router, so any phone or laptop in Wi-Fi range can browse it by its IP address.

    Even without the doomsday angle, it's a good homelab project: Linux, Docker, local networking and storage planning in one box.
4. **A practice Windows domain.** Microsoft offers free, time-limited evaluation copies of Windows Server through its Evaluation Center. Set up Active Directory in a VM, create users and groups, join a Windows VM to the domain, and reset a "user's" password, just as help desk staff do.
5. **A virtual network in Packet Tracer:** build a small office network with a router, switch, DHCP and Wi-Fi, then break it and troubleshoot it (Section 9).
6. **A backup plan:** set up automatic backups for one PC, then prove you can restore a file. Backup and recovery is on A+ Core 2.

11.4 What to practice inside the lab

Zachs Tech Turf, a creator who says he has hired people both in a previous IT job and for his own business, answered a question about teenagers running homelabs in a short Facebook reel. His advice works at any age: real skills you can show set you apart from applicants who only have a degree or a certificate, and all you need to start is a few VMs in VirtualBox on whatever computer you have. In his order:

1. **Linux,** any distribution. The goal is being at home in the terminal, so any other distribution feels familiar.
2. **Windows Server,** because companies still rely on it for Active Directory and for managing Microsoft 365 users.
3. **Networking between your VMs:** set up DHCP, plan subnets, try VLANs, and run your own DNS.
4. **Self-hosted apps,** such as Plex or Jellyfin (project 1 above).
5. **Docker,** since many companies package and test software in containers.

That list lines up with the official material: A+ Core 1 includes networking and virtualization, Network+ covers subnets, VLANs, DHCP and DNS in depth, and Microsoft's free Windows Server evaluation (project 4) and Docker's free Get started guide cover the rest. His one other piece of advice for younger readers: don't get arrested. Keep everything inside your own lab (Section 10).

Be able to explain it. Whatever you build, keep a simple diagram and notes of your setup, and be ready to walk an interviewer through how it fits together and what broke along the way.

12. Resume and Job Search

- **Lead with what you can do.** List your certifications, then a **Projects** or **Homelab** section with specific tasks: "Set up Active Directory in a virtual lab with 20 users and group policies", not "Familiar with Windows".
- **Customer service experience counts.** Retail, hospitality and call center jobs show you can stay calm with frustrated people, which is half of help desk work.
- **Search more than one title:** help desk, IT support, service desk, desktop support, technical support and IT technician all overlap.
- **Apply before you feel ready.** Postings often list more than they need. If you meet most of it, apply.
- **Prepare for practical questions:** "A user can't print. What do you do?" Walk through your steps out loud: check the obvious first, gather information, test one thing at a time, and document what you did.

The Web Developer Careers guide's resume and applications section covers resumes, cover letters and finding postings early, and its advice on keeping going through a long search applies to any field. For interviews, Coding Practice and Interview Prep explains the STAR method for behavioral questions like "Tell me about a time you dealt with a difficult customer."

13. Common Mistakes

- **Studying for a retired exam.** Check that your book, course and practice tests match the exam code you'll book (Sections 5 and 7).
- **Collecting certifications without practice.** One certificate plus a homelab you can explain beats three certificates and nothing to show.
- **Paying for every resource.** Professor Messer's videos, Microsoft Learn and Cisco Networking Academy are free. Pay only for what fills a real gap.
- **Using brain dumps.** They break exam rules, and they don't teach you the job.
- **Practicing on real systems.** Use VMs and your own spare hardware, not a work or school PC, and never someone else's network.
- **Skipping the people skills.** Writing clear ticket notes and explaining fixes simply are what get help desk staff promoted.

14. Sources

Certificates and exams

- Coursera, Google IT Support Professional Certificate
- Grow with Google, Google IT Support Certificate
- CompTIA: A+, A+ Core 1 V15, A+ Core 2 V15, Network+, Security+, Security+ V7, Security+ V8, Linux+ and Earn continuing education units
- Cisco: CCST IT Support exam and CCNA
- Microsoft Learn: Endpoint Administrator Associate and Azure Fundamentals
- AWS, Certified Cloud Practitioner
- PeopleCert, ITIL 4 Foundation

Careers

- US Bureau of Labor Statistics, Occupational Outlook Handbook: Computer Support Specialists
- roadmap.sh

Study resources

- Professor Messer: A+ 220-1201, A+ 220-1202, Network+ N10-009 and Security+ SY0-701
- Cisco Networking Academy: home and Getting Started with Cisco Packet Tracer
- Microsoft Learn training
- Linux Journey and Linux Foundation, Introduction to Linux
- TryHackMe pricing, Hack The Box labs and OverTheWire: Bandit

Hardware

- Kingston: Desktop and notebook memory support and Memory population rules
- Intel, Extreme Memory Profile (XMP)

Homelab

- Microsoft Learn: Install Hyper-V, Windows Sandbox and Install Windows Sandbox
- VirtualBox
- Jellyfin: installation, container install, movie naming, TV naming, hardware acceleration and client downloads
- Plex: naming guides for movies and TV shows
- Plex: Requirements for remote playback of personal media and Remote Watch Pass overview
- Plex, Using Hardware-Accelerated Streaming (last modified October 27, 2026), and the pms-docker README (hardware transcoding in Docker)
- NVIDIA: Video Encode and Decode GPU Support Matrix (concurrent sessions) and Container Toolkit: specialized configurations for Docker
- Community tips on Plex hardware transcoding: user forums and reviews, including XDA
- Backblaze, The 3-2-1 Backup Strategy
- US Copyright Office, Section 1201 exemptions
- Home Assistant
- ChannelNews, IFA 2026: UGREEN Enters Smart Home Market With Local AI Ecosystem
- Project NOMAD and its GitHub repository
- Docker, Get started

Videos

- OCmodshop, Here's Why Your RAM Upgrade Backfired
- Keon Bowman, ssh2john in 60 Seconds
- Madeline and Stephen, How To Make Your Own Streaming Service At Home!
- Salem Techsperts, Ugreen HomeAgent is a Smart Home Hub That YOU Own
- The Modern Rogue, What's In Project NOMAD: The Off-Grid Internet for the Apocalypse?
- Zachs Tech Turf, a short Facebook reel on teenagers running homelabs

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Latest version: github.com/ImmuneMoon/IT-Support-Careers-and-Certifications

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